3. RADIOACTIVITY AND THE BUILDING BLOCKS OF THE NUCLEUS

I. RADIOACTIVITY

When investigating the internal physical properties of any system, we must endeavour, on the one hand, to discover its effects on the outside world and, on the other hand, try to approach it in some way that will show us how it behaves during this process. In certain cases, it becomes necessary to dissect it into its component parts, by means of interference from outside. This principle applies to the atomic nucleus, too. Thus there arises the question whether there are any nuclear phenomena which furnish us with the desired knowledge of their internal structure without our having to resort to such interference. The radioactivity (a phenomenon already mentioned) of certain heavy elements is actually such a phenomenon. For this reason, let us discuss it first.

We have already discussed the fact that in radioactivity three distinct types of radiation can be observed: alpha, beta and gamma radiation. A few years after the first observation of radioactivity, Rutherford and Soddy made the discovery, of decisive importance for the development of atomic theory, that the emission of alpha and beta rays is associated with a transmutation of the chemical elements. An atom which has emitted either of these radiations is no longer an atom of the same original element.

This discovery was of paramount importance for atomic theory. The old notion of atoms had now to be discarded; the atoms of chemistry were, evidently, no longer the ultimate, indivisible building blocks of matter. To be sure, it was still not possible to change one element into another by chemical means, yet the existence of a natural process producing this very result appeared to be a certainty. The hopes and aspirations of the alchemists of past ages were thus given new life. For if in certain cases, nature itself effected a transmutation of the elements, it was bound to be possible to perform this process artificially, once the proper tools were discovered. It was bound to be possible, theoretically at least, to turn mercury into gold!

In view of our present knowledge, in the first place that the alpha and beta particles carry electric charges, and secondly, that the chemical properties of the atom depend on the number of elementary charges of electricity on its nucleus, this discovery of Rutherford and Soddy is easily understandable to-day. The alpha and beta particles originate in the nucleus of the atom, and not in its outer structure. The alpha particles are helium nuclei, identified as such